

Homework 11

1. How many cases and how many variables are in the penguins dataframe?

342 cases and 8 variables

2. How many cases are there after we drop the NA's for the body mass variable?

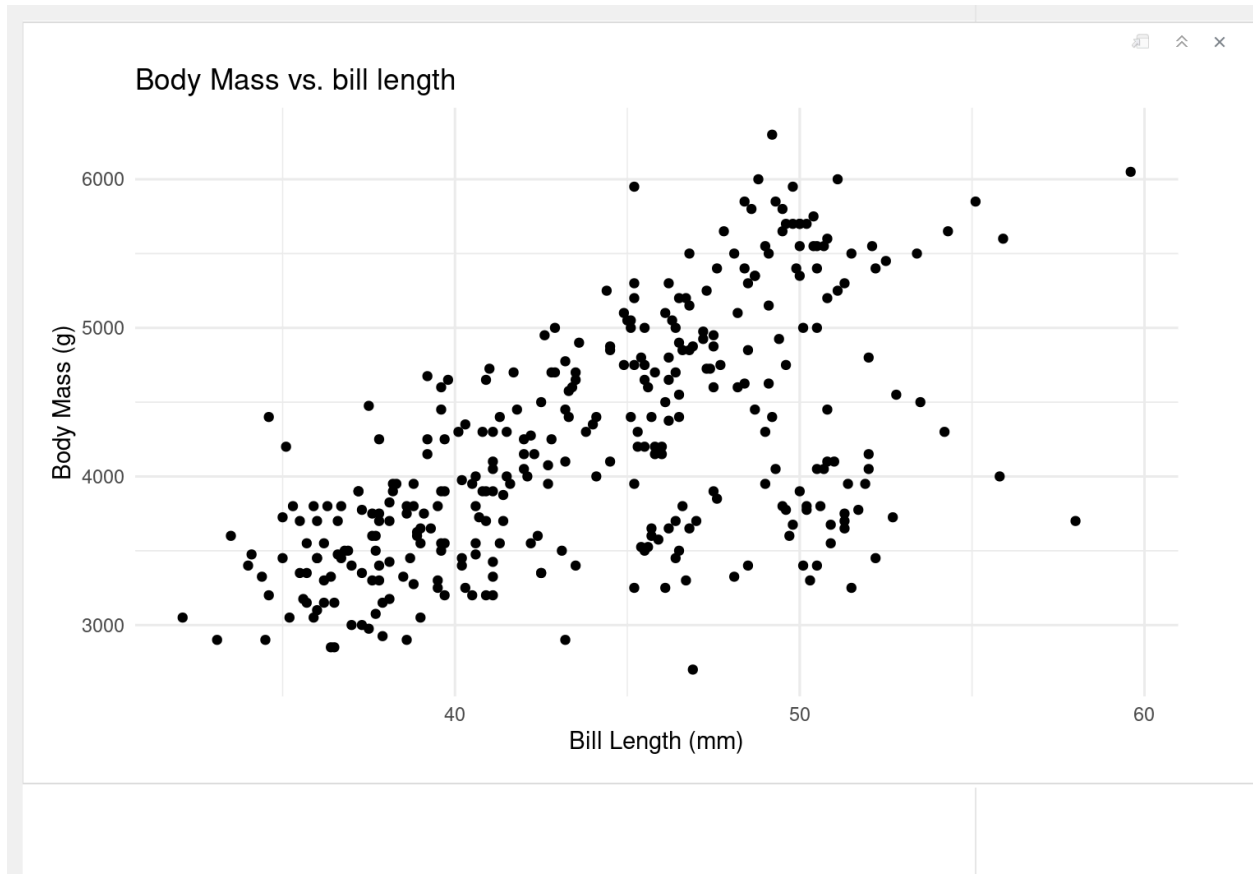
342 cases and 8 variables

*I got the same result both times I checked not sure if this is intended or an error in the dataframe.

3. Throughout, I am interested in making predictions of a penguin's body mass. Does this mean that body mass will be the explanatory variable or the response variable?

Since we are predicting body mass, it will be the response variable

4. Create a graph that would be appropriate for predicting the body mass (in grams) using the bill length (in millimeters). Paste a copy of your graph here. *(Guiding questions: What are the variable types? Which sort of graph is appropriate here? Which variable should go on which axis, given your answer for Question 3?)*

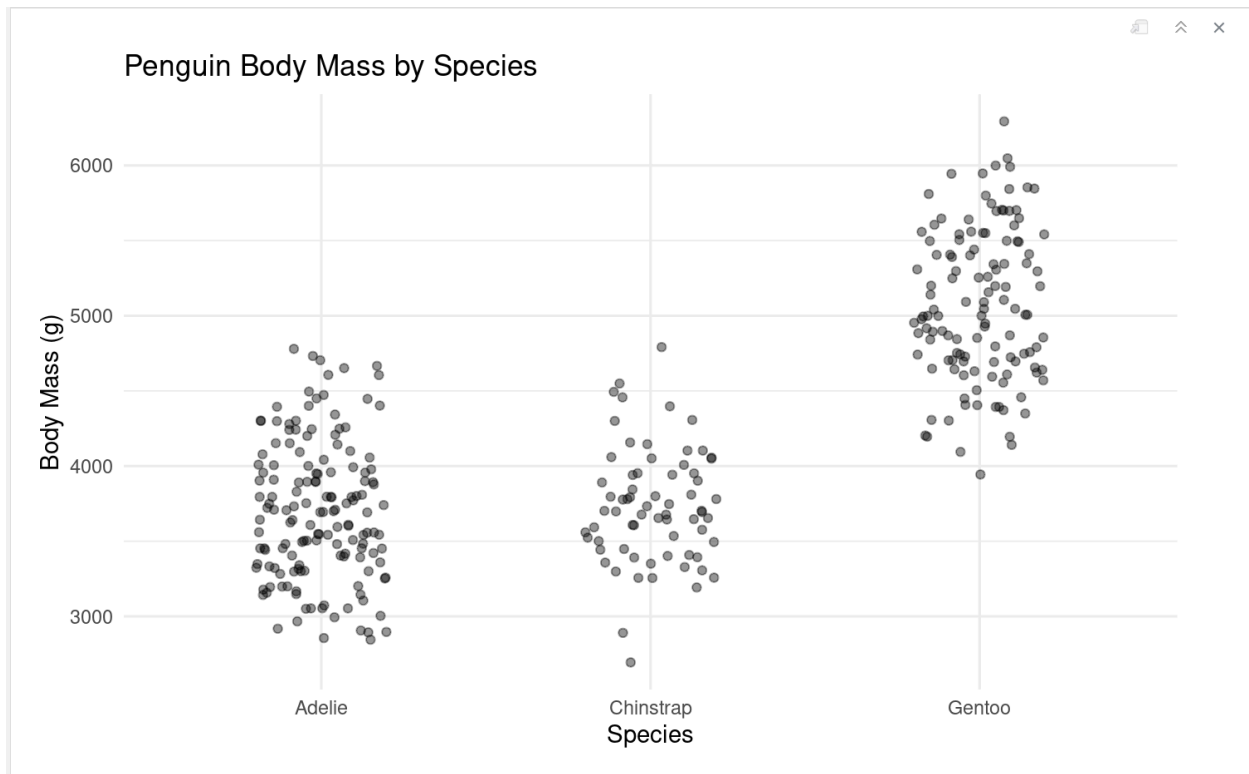


Scatter plots should be used here because both variables are numeric, and bill length (explanatory variable) should go on the x-axis.

5. Does it look to you as if there is an association between these variables? (i.e. do you think bill length will be a reasonable predictor of body mass?) A one-sentence answer is fine here.

Yes, the scatter plot suggests a moderately strong positive association between bill length and body mass. As bill length increases, body mass also tends to increase.

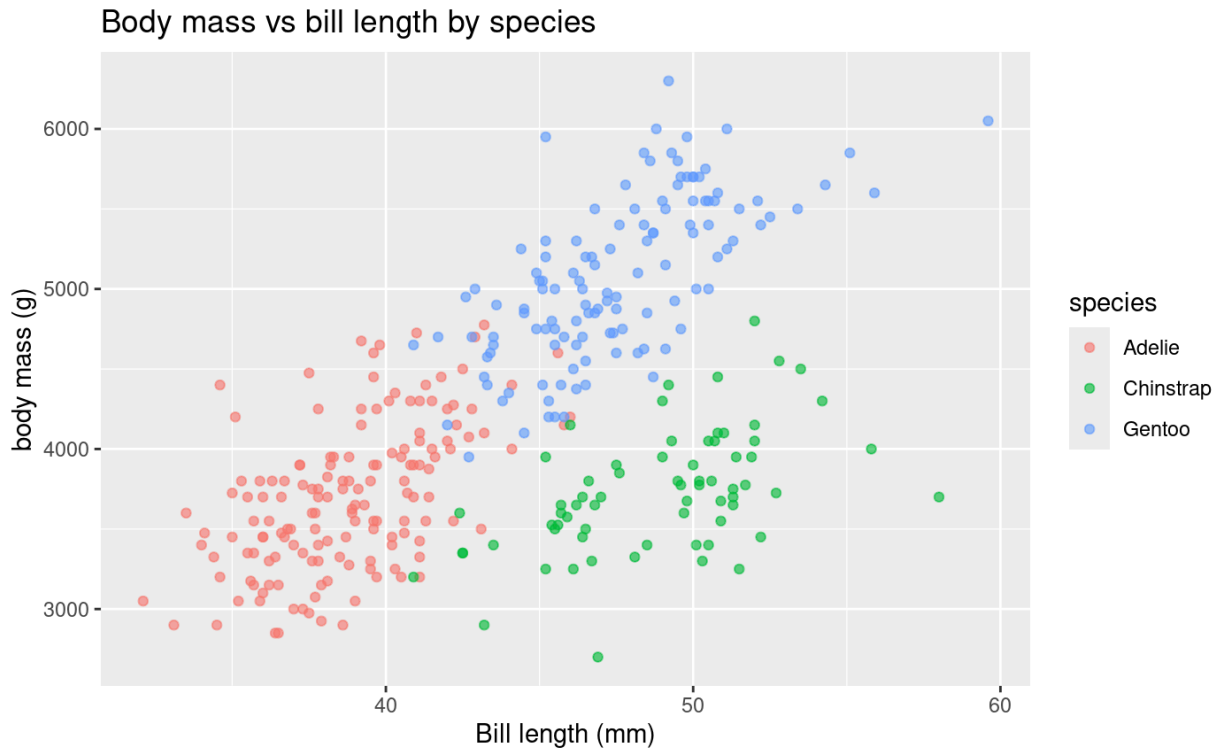
6. Create a graph that would be appropriate for predicting a penguin's body mass knowing only its species. Paste your graph here. *You may wish to look up the work we did on [Frederick_Houses_2](#).*



7. Do you think that there is an association between species and body mass? (One sentence is fine for the moment.)

Yes, there is a clear association between species and body mass. Gentoo penguins tend to have a higher body mass than Adelie and Chinstrap penguins, which suggests that species is a factor in predicting body mass of penguins.

8. Create a graph that would be appropriate for visualizing the body masses of the penguins as predicted by BOTH bill length and species. *You may wish to see how we did this kind of thing in the Frederick Houses Part 3 lab.*



9. Create a model for predicting the body masses of penguins from their bill lengths and species. Write the equation of the model, write sentences of interpretations for each of the coefficients. Make a graph that includes your model lines.

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lm(formula = body_mass_g ~ bill_length_mm + species, data = penguins)
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Residuals:

	Min	1Q	Median	3Q	Max
Residuals	-871.21	-246.17	5.29	213.13	1084.73

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	153.740	268.901	0.572	0.568
bill_length_mm	91.436	6.887	13.276	< 2e-16 ***
speciesChinstrap	-885.812	88.250	-10.038	< 2e-16 ***
speciesGentoo	578.629	75.362	7.678	1.76e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 375.3 on 338 degrees of freedom

Multiple R-squared: 0.7829, Adjusted R-squared: 0.781

F-statistic: 406.3 on 3 and 338 DF, p-value: < 2.2e-16

Body Mass_{hat} = 153.74 + 91.436 * Bill Length - 885.812 * Chinstrap + 578.629 * Gentoo

Note for using the equation:

Chinstrap = 1 if the penguin is Chinstrap, otherwise 0

Gentoo = 1 if the penguin is Gentoo, otherwise 0

The intercept (153.74) represents the predicted body mass when bill length is 0 mm and species is Adelie. The bill_length_mm (91.436) indicates that for every 1 mm increase in bill length, body mass increases by 91.436 g. Additionally the speciesChinstrap (-885.812) means that Chinstrap penguins weigh, on average, 885.812 g less than Adelie penguins, controlling for bill length. Finally, the speciesGentoo (578.629) means that Gentoo penguins weigh, on average 578.629 g more than Adelie penguins, controlling for bill length.

Graph with regression lines:

Body mass vs bill length by species

